Deploy

app
Data Upload
Data Profile
Business Dashboard
AI Insights
Ask Your Data
Forecasting
Executive Report

📊 InsightIQ AI

Autonomous Business Intelligence & Decision Support Platform

Transform raw business data into AI-powered insights, intelligent dashboards, predictive forecasts, and executive reports.

🚀 Platform Highlights

📈 Interactive Dashboards

- Dynamic KPI Cards
- Interactive Charts
- Business Performance Tracking

🧠 AI Business Analysis

- Executive Insights
- Trend Detection
- Business Recommendations

🔮 Predictive Analytics

- Forecast Future Trends
- AI Executive Reports
- Decision Support

✨ Why InsightIQ AI?

- ✅ Supports Multiple Business Datasets
- ✅ Automatic Semantic Column Detection
- ✅ Intelligent KPI Generation
- ✅ AI-Powered Business Insights
- ✅ Interactive Business Dashboards
- ✅ Predictive Forecasting
- ✅ Executive Decision Support
- ✅ Minimal Manual Configuration

📂 Explore the Platform

📂 Data Upload

Upload CSV or Excel datasets and let InsightIQ AI automatically understand your business data.

📊 Business Dashboard

View KPIs, interactive visualizations, and business trends.

📈 Forecasting

Predict future business performance using machine learning.

🔍 Data Profiling

Explore missing values, statistics, data quality, and dataset structure.

🧠 AI Business Analyst

Generate business insights, identify patterns, and receive recommendations.

📄 Executive Report

Generate AI-powered executive summaries for business leaders.

InsightIQ AI • Version 1.0 | Built with Streamlit • Python • Pandas • Plotly • Scikit-learn • Gemini AI

Deploy

Data Upload

Upload your business dataset.

Supported formats:

- ZIP
- CSV
- Excel (.xlsx)

Choose Dataset

Cleaned...alysis.csv
0.6MB

CSV Uploaded Successfully

Shape: `(2845, 28)`

Columns:

```
[
  0 : "Employee ID"
  1 : "StartDate"
  2 : "Title"
  3 : "BusinessUnit"
  4 : "EmployeeStatus"
  5 : "EmployeeType"
  6 : "PayZone"
  7 : "EmployeeClassificationType"
  8 : "DepartmentType"
  9 : "Division"
  10 : "DOB"
  11 : "State"
  12 : "GenderCode"
  13 : "RaceDesc"
  14 : "MaritalDesc"
  15 : "Performance Score"
  16 : "Current Employee Rating"
  17 : "Survey Date"
  18 : "Engagement Score"
  19 : "Satisfaction Score"
  20 : "Work-Life Balance Score"
  21 : "Training Date"
  22 : "Training Program Name"
  23 : "Training Type"
  24 : "Training Outcome"
  25 : "Training Duration(Days)"
  26 : "Training Cost"
  27 : "Age"
]
```

	Employee ID	StartDate	Title	BusinessUnit	EmployeeStatus	EmployeeType	PayZ
0	3427	20-Sep-19	Production Technician I	CCDR	Active	Contract	Zone
1	3428	11-Feb-23	Production Technician I	EW	Active	Contract	Zone
2	3429	10-Dec-18	Area Sales Manager	PL	Active	Full-Time	Zone
3	3430	21-Jun-21	Area Sales Manager	CCDR	Active	Contract	Zone
4	3431	29-Jun-19	Area Sales Manager	TNS	Active	Contract	Zone

JSON Copy

Data Profile

Dataset Understanding


```
{
  "type" : "hr"
  "date_columns" : []
  "numeric_columns" : [
    0 : "Employee ID"
    1 : "Current Employee Rating"
    2 : "Engagement Score"
    3 : "Satisfaction Score"
    4 : "Work-Life Balance Score"
    5 : "Training Duration(Days)"
    6 : "Training Cost"
    7 : "Age"
  ]
  "categorical_columns" : [
    0 : "StartDate"
    1 : "Title"
    2 : "BusinessUnit"
    3 : "EmployeeStatus"
    4 : "EmployeeType"
    5 : "PayZone"
    6 : "EmployeeClassificationType"
    7 : "DepartmentType"
    8 : "Division"
    9 : "Age"
    10 : "State"
    11 : "GenderCode"
    12 : "RaceDesc"
    13 : "MaritalDesc"
    14 : "Performance Score"
    15 : "Survey Date"
    16 : "Training Date"
    17 : "Training Program Name"
    18 : "Training Type"
    19 : "Training Outcome"
  ]
}
```


Semantic Model


```
{
  "date" : "StartDate"
  "region" : "State"
  "price" : "Training Cost"
  "employee" : "Employee ID"
  "department" : "DepartmentType"
  "salary" : "PayZone"
  "status" : "EmployeeStatus"
}
```


Generated Metrics


```
{
  "Total Employees" : 2845
  "Departments" : 6
}
```


Dataset Intelligence


```
{
  "rows" : 2845
  "columns" : 28
  "numeric_columns" : [
    0 : "Employee ID"
    1 : "Current Employee Rating"
    2 : "Engagement Score"
    3 : "Satisfaction Score"
    4 : "Work-Life Balance Score"
    5 : "Training Duration(Days)"
    6 : "Training Cost"
    7 : "Age"
  ]
  "categorical_columns" : [
    0 : "StartDate"
    1 : "Title"
    2 : "BusinessUnit"
    3 : "EmployeeStatus"
    4 : "EmployeeType"
    5 : "PayZone"
    6 : "EmployeeClassificationType"
    7 : "DepartmentType"
    8 : "Division"
    9 : "Age"
    10 : "State"
    11 : "GenderCode"
    12 : "RaceDesc"
    13 : "MaritalDesc"
    14 : "Performance Score"
    15 : "Survey Date"
    16 : "Training Date"
    17 : "Training Program Name"
    18 : "Training Type"
    19 : "Training Outcome"
  ]
  "date_columns" : [
    0 : "StartDate"
    1 : "Survey Date"
    2 : "Training Date"
  ]
  "binary_columns" : [
    0 : "EmployeeStatus"
    1 : "GenderCode"
    2 : "Training Type"
  ]
  "id_columns" : [
    0 : "Employee ID"
  ]
  "measure_columns" : [
    0 : "Current Employee Rating"
    1 : "Engagement Score"
    2 : "Satisfaction Score"
    3 : "Work-Life Balance Score"
    4 : "Training Duration(Days)"
    5 : "Age"
  ]
  "dimension_columns" : [
    0 : "Title"
    1 : "BusinessUnit"
    2 : "EmployeeStatus"
    3 : "EmployeeType"
    4 : "PayZone"
    5 : "EmployeeClassificationType"
    6 : "DepartmentType"
    7 : "Division"
    8 : "State"
    9 : "GenderCode"
    10 : "RaceDesc"
    11 : "MaritalDesc"
    12 : "Performance Score"
    13 : "Training Program Name"
    14 : "Training Type"
    15 : "Training Outcome"
  ]
}
```


Dataset loaded successfully

Preview

	Employee ID	StartDate	Title	BusinessUnit	EmployeeStatus	EmployeeType	PayZ
0	2407	25-Sep-18	Production Technician I	CCDH	Active	Contract	Zone
1	2408	11-Feb-15	Production Technician I	EW	Active	Contract	Zone
2	2409	14-Dec-18	Area Sales Manager	PL	Active	Full Time	Zone
3	2410	11-Jun-11	Area Sales Manager	CCDH	Active	Contract	Zone
4	2411	25-Jun-18	Area Sales Manager	TKS	Active	Contract	Zone

Dataset Profile

Rows	Columns	Missing Values	Duplicates
2845	28	0	0

Memory (MB)	Numeric Columns	Categorical Columns	Date Columns
1.11	8	20	3

Column Information

	Column	Data Type	Missing Values	Unique Values
EmployeeID	Employee ID	int64	0	2845
StartDate	StartDate	str	0	1473
Title	Title	str	0	32
BusinessUnit	BusinessUnit	str	0	10
EmployeeStatus	EmployeeStatus	str	0	3
EmployeeType	EmployeeType	str	0	3
PayZone	PayZone	str	0	3
EmployeeClassificationType	EmployeeClassificationType	str	0	3
DepartmentType	DepartmentType	str	0	6
Division	Division	str	0	15

Dataset Summary

	Employee ID	StartDate	Title	BusinessUnit	EmployeeStatus	EmployeeType	PayZone	EmployeeID
count	2845	2845	2845	2845	2845	2845	2845	2845
unique	Nan	1473	32	10	3	3	3	3
top	Nan	04-Mar-13	Produc	SVL	Active	Full Time	Zone 6	Temporary
freq	Nan	7	1241	251	2438	897	1111	888
mean	2478.2223	Nan	Nan	Nan	Nan	Nan	Nan	Nan
std	848.4464	Nan	Nan	Nan	Nan	Nan	Nan	Nan
min	2001	Nan	Nan	Nan	Nan	Nan	Nan	Nan
25%	2735	Nan	Nan	Nan	Nan	Nan	Nan	Nan
50%	3446	Nan	Nan	Nan	Nan	Nan	Nan	Nan
75%	3537	Nan	Nan	Nan	Nan	Nan	Nan	Nan

Missing Values

No missing values found

Business Dashboard
Dashboard Data Check
Business KPIs
2845
2845
6
3.03
2.94
2.99
2458
5
0
Business Visualizations

Deploy

💬 Ask Your Data

Ask a question about your data

advices for the hr to improve

As an expert Business Intelligence Analyst, I've reviewed your dataset to identify key areas where HR can focus to drive improvement. My advice is structured around several critical aspects of the employee lifecycle and experience.

Here are some data-driven advices for HR to improve, based on the provided columns:

1. Enhance Employee Well-being and Engagement

Insights from: 'Engagement Score', 'Satisfaction Score', 'Work-Life Balance Score', 'Performance Score', 'Current Employee Rating'

- **Identify and Address Low Scores:** Calculate the average 'Engagement Score', 'Satisfaction Score', and 'Work-Life Balance Score' across the organization. HR should then segment these scores by 'DepartmentType', 'Division', 'EmployeeType', 'PayZone', 'Age', 'GenderCode', and 'RaceDesc'.
  - **Actionable Advice:** Investigate groups (e.g., specific departments, divisions, or demographic segments) that consistently show lower scores in any of these areas. This might indicate issues with workload, management style, recognition, or inclusive practices. HR should conduct pulse surveys or focus groups to understand the root causes and implement targeted interventions (e.g., flexible work policies, mental health support, team-building initiatives, workload re-evaluation).
- **Link Well-being to Performance:** Analyze the correlation between 'Engagement Score', 'Satisfaction Score', 'Work-Life Balance Score' and 'Performance Score'/'Current Employee Rating'.
  - **Actionable Advice:** If there's a noticeable link, HR should emphasize that fostering a positive work environment and supporting employee well-being directly contributes to better performance. Initiatives to boost these scores could be framed as performance improvement strategies.

2. Optimize Training Programs and Investment

Insights from: 'Training Program Name', 'Training Type', 'Training Outcome', 'Training Duration(Days)', 'Training Cost', 'Performance Score'

- **Evaluate Training Effectiveness:** Calculate the success rate (percentage of 'Completed' outcomes) for each 'Training Program Name' and 'Training Type'. Also, analyze the occurrence of 'Failed' or 'Incomplete' outcomes.
  - **Actionable Advice:** Review training programs with consistently high 'Failed' or 'Incomplete' outcomes. This could indicate issues with curriculum design, delivery method, trainer effectiveness, or participant selection. HR should consider post-training assessments, participant feedback, and pre-requisites to improve outcomes.
- **Assess Return on Training Investment:** Correlate 'Training Cost' and 'Training Duration(Days)' with 'Training Outcome' and subsequent 'Performance Score' / 'Current Employee Rating'.
  - **Actionable Advice:** Identify if high-cost or long-duration training programs are yielding proportionate positive outcomes or performance improvements. If not, HR should re-evaluate the investment, potentially seeking more cost-effective or shorter, high-impact alternatives. Similarly, identify successful, cost-effective programs to replicate.
- **Align Training with Performance:** Analyze if employees who completed specific training programs (especially 'Completed' outcomes) show an improvement in their 'Performance Score' or 'Current Employee Rating' in subsequent periods.
  - **Actionable Advice:** Prioritize and promote training programs that demonstrably lead to higher performance or ratings. This also allows HR to demonstrate the tangible value of training to the business.

3. Review Employee Classification and Business Unit Dynamics

Insights from: 'EmployeeType', 'EmployeeClassificationType', 'BusinessUnit', 'DepartmentType', all scores and outcomes.

- **Segmented Employee Experience:** Compare engagement, satisfaction, work-life balance, and performance scores across different 'EmployeeType' (e.g., Contract vs. Full-Time) and 'EmployeeClassificationType' (e.g., Temporary, Part-Time, Full-Time). Do the 'Contract' or 'Temporary' employees experience significantly different outcomes?
  - **Actionable Advice:** Understand the unique challenges and needs of different employee classifications. HR might need tailored support, communication strategies, or benefit packages to ensure equity and maximize engagement across all employee types, especially if disparities are observed.
- **Business Unit & Department Benchmarking:** Benchmark scores and training outcomes across different 'BusinessUnit' and 'DepartmentType'.
  - **Actionable Advice:** Identify best practices from high-performing business units or departments with high employee scores and successful training outcomes. Conversely, investigate underperforming units or departments to identify specific issues and implement targeted support or development programs.

4. Promote Diversity, Equity, and Inclusion (DEI)

Insights from: 'GenderCode', 'RaceDesc', 'MaritalDesc', 'Age', 'PayZone', all scores and outcomes.

- **Equity in Experience and Opportunities:** Analyze if there are significant disparities in 'Engagement Score', 'Satisfaction Score', 'Work-Life Balance Score', 'Performance Score', 'Current Employee Rating', 'Training Outcome', or 'PayZone' across different 'GenderCode', 'RaceDesc', 'MaritalDesc', or 'Age' groups.
  - **Actionable Advice:** If any group consistently shows lower scores or less favorable outcomes, HR should investigate potential systemic biases or unmet needs. This requires a deeper dive into pay equity, promotion opportunities, access to development, and overall inclusive culture. Implement targeted DEI initiatives and unconscious bias training.

By systematically analyzing these metrics and patterns, HR can make data-informed decisions to improve employee experience, optimize training investments, and ultimately contribute more strategically to the organization's success.

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AI Insights

Gemini Business Insights

Here's a complete business analysis based on the provided HR dataset and KPIs:

HR Dataset Business Analysis

Dataset Overview: The dataset contains 2845 records with 28 columns, covering employee demographics, employment details, performance metrics, survey scores, and training information. Notably, there are no missing values, indicating high data quality.

Key Performance Indicators (KPIs):

- **Total Records & Employees:** 2845
- **Active Employees:** 2458 (approx. 86.4% of total)
- **Departments:** 6 distinct departments
- **Training Programs:** 5 distinct training programs
- **Average Satisfaction Score:** 3.03
- **Average Engagement Score:** 2.94
- **Average Work-Life Balance Score:** 2.99

---

1. Key Business Insights

- **Workforce Size and Status:** The organization has a substantial workforce of 2845 employees. A significant majority (2458 employees, or approximately 86.4%) are currently active, indicating a relatively stable operational workforce.
- **Organizational Structure:** The company operates across 6 distinct departments, suggesting a moderately diversified or specialized organizational structure. Additionally, 5 different training programs are in place, indicating an investment in employee development.
- **Employee Sentiment (Overall Neutrality):** The average scores for Satisfaction (3.03), Engagement (2.94), and Work-Life Balance (2.99) are all consistently around the neutral mark (assuming a scale, likely 1-5 where 3 is neutral). This suggests that employees generally feel neither strongly positive nor strongly negative across these key sentiment areas.
- **Data Quality:** The absence of missing values across all columns is a significant positive insight, ensuring the reliability and completeness of the data for analysis.

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2. Important Trends

- **Time-Based Trends:** Based on the provided KPIs and dataset analysis, specific time-based trends in employee sentiment, performance, or attrition **cannot be identified**. While date columns (`StartDate` , `Survey Date` , `Training Date`) exist in the dataset, the aggregated KPIs do not provide a temporal view. Further analysis would be required to observe changes over time.
- **Cross-Sectional Observations (from available data types):**
  - The dataset contains various categorical dimensions such as `DepartmentType` , `Division` , `Title` , `EmployeeType` , `PayZone` , `EmployeeClassificationType` , `State` , `GenderCode` , `RaceDesc` , `MaritalDesc` . Analysis across these dimensions (e.g., satisfaction scores by department or age group) *could* reveal important trends, but these are not provided in the current KPIs.
  - The presence of `Training Program Name` and `Training Outcome` suggests that detailed analysis of training effectiveness across different programs is possible.

---

3. Risks or Problems

- **Mediocre Employee Sentiment:** The average scores for Engagement, Satisfaction, and Work-Life Balance hovering around '3' indicate a risk of widespread neutrality rather than strong positive sentiment.
  - **Possible Interpretation:** This 'average' sentiment could mask pockets of dissatisfaction or disengagement within specific departments, divisions, or employee groups. It might also suggest a lack of compelling initiatives to significantly boost employee morale and loyalty.
- **Inactive Employee Population:** With 387 employees (approximately 13.6% of the total records) not being active, there is a substantial number of individuals who are no longer contributing to the organization.
  - **Possible Interpretation:** Without knowing the reasons (`EmployeeStatus` breakdown), this could represent high turnover, a significant number of employees on leave, or a backlog in HR processes. High turnover can lead to increased recruitment costs and loss of institutional knowledge.
- **Training Program Effectiveness:** The existence of `Training Outcome` and `Training Cost` columns implies that training effectiveness is a measurable aspect. The sample data shows "Failed" and "Incomplete" outcomes, which, if prevalent across the organization, represents a direct waste of `Training Cost` and potentially a skill gap.
- **Potential for Undiagnosed Issues:** The current average scores for sentiment, while not alarming, suggest a lack of clear strengths. Without deeper segmentation by demographics, roles, or performance, critical issues affecting specific employee groups might remain undiagnosed.

---

4. Business Recommendations

- **Deep Dive into Employee Sentiment:**
  - Conduct a detailed analysis of `Engagement Score` , `Satisfaction Score` , and `Work-Life Balance Score` broken down by `DepartmentType` , `Division` , `Title` , `Age` , `GenderCode` , and `EmployeeClassificationType` .
  - Identify specific groups or departments that show significantly lower (or higher) scores than the average. This will enable targeted interventions rather than broad, less effective initiatives.
  - **Recommendation:** Implement strategies to elevate overall employee experience, aiming to move average scores above the neutral mark of '3'. This could involve surveying employees for qualitative feedback on areas for improvement.
- **Analyze Inactive Employee Data:**
  - Investigate the specific reasons behind the `EmployeeStatus` for the 387 inactive employees. Categorize these statuses (e.g., voluntary attrition, involuntary separation, leave of absence) to understand the underlying causes.
  - **Recommendation:** If high attrition is identified, develop retention strategies. If other issues (e.g., prolonged leaves) are prevalent, review related policies and support systems.
- **Optimize Training Investment:**
  - Analyze `Training Outcome` in relation to `Training Program Name` , `Training Type` , and `Training Cost` .
  - Identify programs with low success rates ("Failed," "Incomplete") or high costs that do not yield desired outcomes.
  - **Recommendation:** Evaluate and refine or discontinue underperforming training programs to ensure `Training Cost` translates into effective skill development and positive `Training Outcome` . Consider correlating training outcomes with `Performance Score` to assess business impact.
- **Leverage Performance Data for Holistic Insights:**
  - Correlate `Performance Score` and `Current Employee Rating` with `Engagement Score` , `Satisfaction Score` , and `Work-Life Balance Score` .
  - **Recommendation:** Understand if there's a direct relationship between employee sentiment and performance. This can highlight whether improving employee experience directly impacts performance, guiding future HR strategies.
- **Implement Time-Series Analysis:**
  - Utilize `StartDate` , `Survey Date` , and `Training Date` to track trends over time for all relevant KPIs (e.g., how average satisfaction changes quarter-over-quarter, or the evolution of training outcomes).
  - **Recommendation:** Establish regular reporting cycles for these time-based metrics to proactively identify emerging issues or measure the impact of HR initiatives.

Deploy

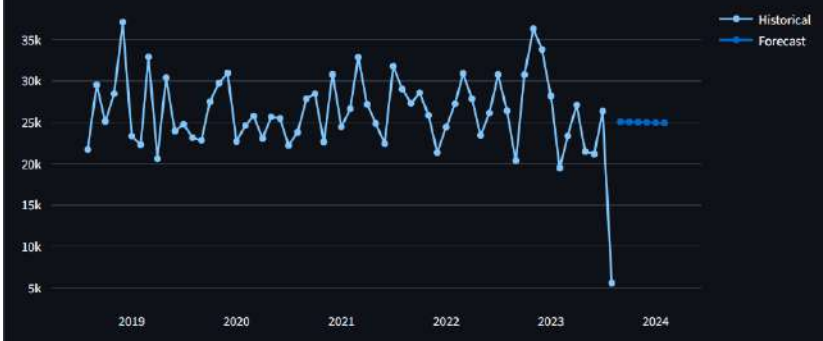
📈 Business Forecasting

📊 Forecast Capability

Forecast Available: Training Cost

Detected Dataset: HR

📈 Training Cost Forecast



📌 Forecast Summary

Current Value	Future Prediction	Expected Change
5593.43	24929.61	345.69%

Forecast trend: Increasing

🤖 AI Forecast Explanation

Here is a short business forecast explanation for TrainingCost:

1. Forecast Observation The forecast for TrainingCost shows a substantial increase. The historical last recorded value was 5593.43, and the future predicted value is 24929.61. This represents an expected change of 345.69%.

2. Trend Explanation This significant projected rise is consistent with an "Increasing" trend in TrainingCost. The future predicted value is nearly 4.5 times the historical value, indicating a strong upward trajectory for training expenditures.

3. Assumptions and Limitations This forecast relies exclusively on the provided historical and predicted data for TrainingCost. We assume the forecast model is accurate given the provided inputs. No external factors influencing TrainingCost (such as changes in organizational strategy, market conditions, or regulatory shifts) were provided or considered in this explanation.

🔮 Future Prediction

	TimeIndex	TrainingCost	Date
0	61	25089.9985	2023-09-01 00:00:00
1	62	25057.9209	2023-10-01 00:00:00
2	63	25025.8433	2023-11-01 00:00:00
3	64	24993.7657	2023-12-01 00:00:00
4	65	24961.6881	2024-01-01 00:00:00
5	66	24929.6105	2024-02-01 00:00:00

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Executive Business Report

Detected Dataset Type: HR

Business Overview

Total Records	Employees	Departments	Avg Satisfaction
2845	2845	6	3.03

Avg Engagement	Avg Work-Life Balance	Active Employees	Training Programs
2.94	2.99	2458	5

Missing Values
0

Executive Summary

Executive HR Report

Date: October 26, 2023

Prepared For: Business Leadership

Prepared By: Senior Business Intelligence Analyst

---

1. Overall Workforce Performance

The organization maintains a robust set of HR data with **2,845 total employee records** and **0 missing values**, indicating excellent data quality. Currently, **2,458 employees are actively engaged** in the workforce, distributed across **6 distinct departments**.

Employee sentiment, as measured by recent surveys, indicates moderate scores:

- **Average Employee Satisfaction:** 3.03
- **Average Employee Engagement:** 2.94
- **Average Work-Life Balance:** 2.99

The organization is committed to employee development, offering **5 distinct training programs** as part of its learning initiatives.

2. Key Workforce Trends and Patterns

- **Workforce Status Dynamics:** A notable pattern emerges from the difference between total employee records (2,845) and currently active employees (2,458). This suggests a historical component of 387 inactive employee records, which warrants further investigation into the employee lifecycle and potential attrition.
- **Employee Experience Consistency:** The average scores for satisfaction, engagement, and work-life balance consistently hover around the midpoint. This indicates a general baseline for employee experience across the organization, rather than outliers of exceptionally high or low sentiment. Engagement, at 2.94, is marginally lower than satisfaction and work-life balance.
- **Structured Development Approach:** The presence of 5 unique training programs ('Training Program Name') indicates a structured effort to enhance employee skills. Further analysis of 'Training Type', 'Training Outcome', 'Training Duration(Days)', and 'Training Cost' could reveal specific patterns in training effectiveness and resource allocation.
- **Demographic Data Availability:** The dataset provides detailed demographic information including 'Age', 'GenderCode', 'RaceDesc', and 'MaritalDesc'. This rich data can be leveraged for deeper analysis into diversity, equity, and inclusion patterns, and to tailor HR initiatives.
- **Time-Based Analysis Potential:** While specific time-based trends are not calculated in this summary, the presence of 'StartDate', 'Survey Date', and 'Training Date' columns signifies the potential for future time-series analysis to track changes in employee tenure, sentiment, and training participation over time.

3. Major Risks or Concerns

- **Workforce Inactivity/Attrition:** The presence of **387 inactive employee records** (difference between total records and active employees) is a significant concern. Without understanding the specific reasons (e.g., voluntary attrition, retirement, termination), this represents a potential risk to workforce stability, institutional knowledge, and talent pipeline.
- **Moderate Employee Sentiment Scores:** The average scores for **Satisfaction (3.03), Engagement (2.94), and Work-Life Balance (2.99)** are not indicative of high employee morale or optimal work environment. Sustained moderate scores in these critical areas can lead to decreased productivity, higher attrition, and difficulties in attracting top talent in the long term. The slightly lower engagement score (2.94) is particularly noteworthy.
- **Unquantified Training Efficacy:** While training programs are provided, the overall effectiveness and return on investment ('Training Cost' exists but no aggregated financial impact is calculated) remain unquantified in the available KPIs. The 'Training Outcome' column suggests that not all programs may result in successful completion (as seen in sample data, e.g., 'Failed', 'Incomplete'). This poses a risk of inefficient resource allocation if training programs are not yielding desired results.

4. Strategic Recommendations

- **Investigate Inactive Employee Statuses:** Conduct a thorough analysis of the inactive employee records by leveraging 'EmployeeStatus' and 'StartDate' to identify the primary drivers of inactivity/attrition. Develop and implement targeted retention strategies based on these findings.
- **Elevate Employee Experience:** Develop and launch initiatives aimed at improving average employee satisfaction, engagement, and work-life balance scores. This could include leadership training focused on employee support, enhancement of benefits, flexible work options, and recognition programs. Prioritize engagement-boosting initiatives given its slightly lower score.
- **Optimize Learning and Development Programs:** Analyze the 'Training Outcome' data in conjunction with 'Training Cost' and 'Training Duration(Days)' for each 'Training Program Name' to identify strengths and weaknesses. Redesign or refine programs that show lower effectiveness to ensure optimal skill development and resource utilization.
- **Leverage Data for Targeted HR Interventions:** Utilize the rich demographic data ('Age', 'GenderCode', 'RaceDesc', 'MaritalDesc', 'DepartmentType', 'Division') to identify specific employee segments that may benefit from tailored HR support, diversity & inclusion initiatives, or career development pathways.
- **Establish Performance Benchmarks:** Set clear, measurable targets for improvement in average satisfaction, engagement, and work-life balance scores over future periods. Implement regular pulse checks or surveys to monitor progress and adapt strategies as needed.

Training Cost Forecast

Future Prediction

	Date	TrainingCost
0	2023-09-01 00:00:00	25089.9983
1	2023-10-01 00:00:00	25057.3209
2	2023-11-01 00:00:00	25025.6433
3	2023-12-01 00:00:00	24993.7657
4	2024-01-01 00:00:00	24961.6881
5	2024-02-01 00:00:00	24929.6105